

HITL Setup

ArduPlane fmu3-hil + X-Plane 11

Hardware-in-the-Loop (HITL) using a Pixhawk v2 (fmuv3) running custom ArduPlane firmware. X-Plane provides the flight model; the Pixhawk runs the real autopilot code with sensor data injected over MAVLink.

Architecture

The diagram below shows the data flow between X-Plane 11, the MAVLink bridge script, MAVProxy, QGroundControl, and the Pixhawk flight controller.

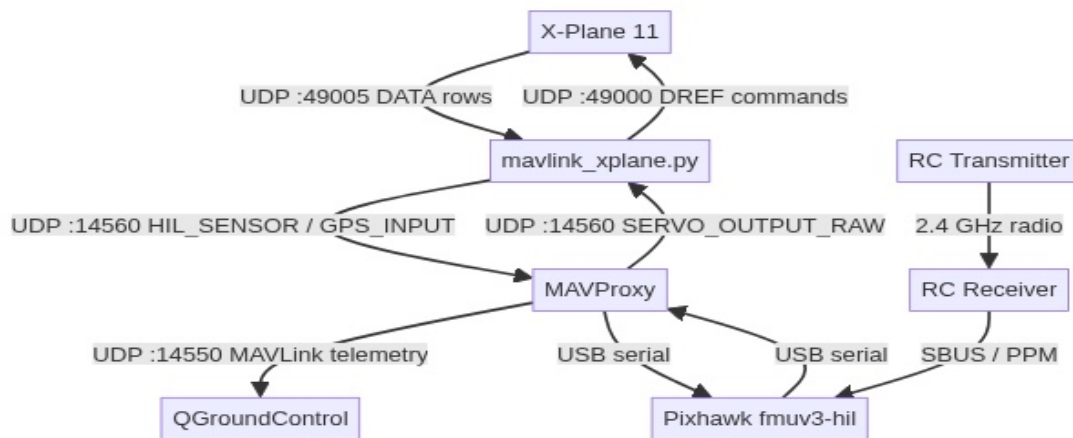
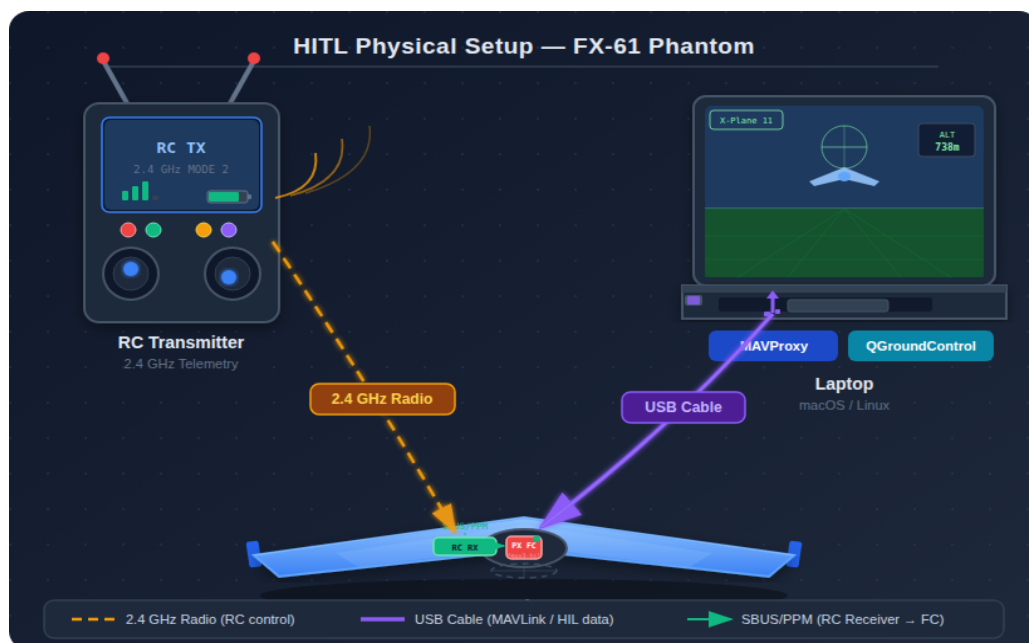


Figure 1 — HITL Architecture Diagram

Physical Setup



Prerequisites

```
pip3 install pymavlink mavproxy
```

Firmware: ArduPlane built for `fmuv3-hil` (includes `HIL_SENSOR` handling and embedded `defaults.parm`). Flash once via Mission Planner or `uploader.py`.

X-Plane 11: running, with a fixed-wing aircraft loaded, sim unpaused. In X-Plane → Settings → Net Connections → Data: send data to IP `127.0.0.1` port `49005`; receive commands on port `49000` (default).

Step 1 — Connect Pixhawk via USB

Plug the Pixhawk into the laptop with a USB cable. Find the serial port:

```
# macOS
ls /dev/tty.usbmodem*

# Linux
ls /dev/ttyACM*
```

Note the port, e.g. `/dev/tty.usbmodem1101`.

Step 2 — Run MAVProxy (serial → UDP mux)

MAVProxy bridges the USB serial link and fans out MAVLink to both QGroundControl and the bridge script.

```
mavproxy.py \
  --master /dev/tty.usbmodem1101 \
  --baudrate 115200 \
  --out udp:127.0.0.1:14550 \
  --out udp:127.0.0.1:14560 \
  --daemon
```

Replace `/dev/tty.usbmodem1101` with your actual port. `--daemon` keeps MAVProxy in the background (omit to keep it in the foreground for debugging). Leave this terminal open / running before starting anything else.

Confirm it connects — you should see heartbeat lines or no error output.

Step 3 — Run QGroundControl

Open **QGroundControl**. It will auto-connect to UDP `:14550`. You should see: vehicle connected (ArduPlane), parameters loaded, and flight mode displayed (e.g. `MANUAL`).

QGC is used for flight planning, mode changes, and parameter tuning. It does **not** need to be started before MAVProxy.

Step 4 — Run mavlink_xplane.py

```
cd /path/to/ardupilot

python3 mavlink_xplane.py \
  --pixhawk udp:127.0.0.1:14560 \
  --xplane-host 127.0.0.1 \
  --xplane-port 49000 \
  --bind-port 49005 \
  --gps-rate 5 \
  --hil-rate 100
```

Add `--debug` to print attitude, sensor, and DREF values every 5 seconds.

Expected startup output

```
[MAV] Connecting to udp:127.0.0.1:14560 ...
[MAV] Heartbeat sysid=1 compid=0
[XP] Listening on :49005 -> ('127.0.0.1', 49000)
[XP] Parking brake SET
[XP] DSEL rows [1, 3, 4, 16, 17, 20, 21]

Running -- Ctrl+C to stop

[XP] X-Plane 11 (build 115501)
[XP] First DREF -> ('127.0.0.1', 49000)
[GPS] lat=-6.900300 lon=107.575200 alt=738.0 m hdg=103.0 deg
[SRV] ail=1500 elev=1500 thr=1100 rud=1500 (50 msg/s)
```

Troubleshooting

Symptom	Likely cause	Fix
[wait] No X-Plane DATA@	X-Plane not sending data	Check Net Connections in X-Plane; unpause sim
[XP] First DREF never prints	No SERVO_OUTPUT_RAW from FC	Check MAVProxy is running; FC is armed or ARMING_REQUIRE=0
Controls don't move in X-Plane	Override DREFs lost	Wait up to 5 s for automatic refresh; reload X-Plane aircraft
EKF not initialising	GPS_INPUT not accepted	Verify GPS1_TYPE=14 in QGC parameters
Heading wrong	Hardware compass active	Verify COMPASS_USE=0,1,2 are set (embedded in defaults.parm)

Step 5 — Run X-Plane 11

1. Open **X-Plane 11** and load a fixed-wing aircraft at the target airport.
2. Configure network output (only needed once — X-Plane saves the setting):

Go to **Settings** → **Net Connections** → **Data**. Under **Send network data output**, enable and set IP address: 127.0.0.1, port: 49005. Under **Accept network data input**, port should be 49000 (default).

3. Enable the required DATA@ rows in X-Plane's **Data Output** panel (**Settings** → **Data Output**). Tick **"Send data over the net"** for each row:

Row #	Name	Fields used
1	Frame rate	sim time (elapsed seconds)
3	Speeds	IAS (knots) → diff pressure for airspeed sensor
4	G-load	body-frame accelerations (x, y, z)
16	Angular velocities	roll/pitch/yaw rate → gyro
17	Pitch, roll, heading	attitude + true heading → GPS yaw
20	Lat, lon, altitude	GPS position
21	Loc, vel, dist	NED velocity → GPS velocity

Rows that are **not** ticked will not be sent and the bridge will silently use stale/zero values for those sensors.

4. **Unpause** the simulation (press P or click the pause button).

Once unparsed, mavlink_xplane.py will start receiving DATA@ packets and inject sensor data into the Pixhawk. You should see [GPS] lines updating every second and the aircraft attitude in QGC matching X-Plane.

Parameter Notes

All required parameters are embedded in defaults.parm and applied automatically on first boot with clean EEPROM. Key values:

Parameter	Value	Purpose
GPS1_TYPE	14	Accept GPS_INPUT MAVLink messages
ARSPD_TYPE	100	SITL airspeed backend (reads HIL_SENSOR diff pressure)
EK3_SRC1_YAW	2	Yaw from GPS_INPUT heading field
COMPASS_USE	0	Disable hardware compass fusion
SCHED_LOOP_RATE	100	Allow 50 Hz SERVO_OUTPUT_RAW stream
BRD_SAFETY_DEFLT	0	Disable safety switch requirement

To reset parameters to defaults (e.g. after flashing new firmware):

```
# In MAVProxy console
param set FORMAT_VERSION 0
reboot
```

Stopping

1. Ctrl+C in the mavlink_xplane.py terminal.
2. Close QGroundControl.
3. Ctrl+C in the MAVProxy terminal (or quit in the MAVProxy console).